

KV-fork

The engine branches the key and value cache and resumes; nothing is re-read.

$$r_k = 0$$



Prefix cache

The cached prefix is reused and only the cue is prefilled again.

$$r_k = 12$$



Black-box API

The whole prefix is resubmitted with every probe.

$$r_k = t_k + 12$$

Only the re-prefill term changes. On GSM8K with Qwen3-8B the same smoothed-confidence policy is roughly cost-neutral in the first regime and wastes 82.3% of the budget in the third (Table 5.6), so an overhead figure without its regime is not comparable.